

Vishwanatha Kanchaboina

Cincinnati, Ohio | vishwanathakanchaboina@gmail.com | [Linked In](#) | (513) 410-5572 | Website: [Portfolio](#) | [GitHub](#)

Skills: Java 8, Kotlin, JavaScript, C, C++, C#, python, Keras, Pandas, OpenCV, NLP, ReactJS, AngularJS, HTML, CSS, Bootstrap, Spring, Spring Boot, Maven, Mockito, HTTP, REST, MySQL, MongoDB, JDBC, Oracle, Apache Kafka, Apache Storm, AWS, Azure, GCP, Docker, OpenShift, Redis, Git

Experience:

Global Quality Corp | (Full-stack Developer):

04/2024 – 08/2024

- Developed a Predictive Maintenance system for Vancouver's sewage using EfficientDet, Mask R-CNN, LightGBM, XGBoost, CatBoost, and RFC with CUDA for defect detection and prediction, reducing training time by 34%. Utilized Apache Kafka, Apache Storm, and Apache Spark with Java to implement real-time data processing and analytics for the ML models, enhancing real-time data ingestion and processing capabilities by 50%.
- Managed a distributed sensor network with ESP32 microcontrollers, ARM-based nodes, and a central x86 root node, using ESP-IDF and DeepVibe nodes. Developed a Linux Socket App for data processing and FFT analysis, interfacing data to cloud services via Spring Boot and Spring Data JPA. Utilized TDEngine PostgreSQL for time-series data storage and visualized in Grafana with custom plugins.
- Designed scalable front-end applications using React, Vue, and Next.js, implementing state management with Redux for responsive design.

Cognizant Technology Solutions

Jnr Software Engineer/ SME

07/2022 – 08/2023

- Conducted in-depth sanity checks for Java-based code coverage and security for **Gemini AI (formerly Bard AI)**, focusing on Large Language Models (LLMs). Enhanced the accuracy of AI-driven outputs by 5% through detailed analysis and quality assurance
- **Engineered:** a scalable Spring Boot-based microservices architecture, integrating Kafka messaging for communication and Java ExecutorService for multi-threading for distributed inventory tracking across **2700+** Kroger warehouses.
- **Developed:** PL/pgSQL procedures on AWS RDS, optimizing inventory databases and accelerating stock queries by 30% with **Redis caching**.
- **Expedited:** Release cycles by achieving 95% code coverage utilizing JUnit and Mockito focused on stock and transaction processing APIs. Load-tested 100,000 concurrent requests utilizing TestNG to ensure production readiness

Programmer Analyst Trainee

09/2021 – 07/2022

- Directed the implementation of an IoT-enabled system across manufacturing lines, utilizing PySpark, Apache Kafka, and Grafana for real-time data analytics and monitoring. This project resulted in a 12% increase in production efficiency and a 3% decrease in energy consumption.
- Orchestrated a comprehensive CI/CD pipeline using Jenkins and Harness, significantly accelerating deployment cycles by 45% and cutting down release-related incidents by 30%, ensuring seamless and reliable software delivery.
- Implemented advanced monitoring and logging solutions using Dynatrace and Splunk, coordinated with Zookeeper for distributed systems management, achieving a 30% reduction in incident response times and ensuring 97% system uptime, thereby enhancing overall system.

Full Stack Intern

02/2021 – 06/2021

- Led and managed a team of 30 full-stack interns, setting goals, providing guidance, and conducting performance evaluations. Developed end-to-end insurance management applications using Spring and Spring Boot for backend services, Maven for build automation, and ReactJS/AngularJS for front-end development in an Agile environment.
- Utilized AWS for cloud storage, Git for version control, and handled network-level testing with tools like Charles and BurpSuite
- Experienced with REST/JSON APIs using Postman, and managed releases through CI/CD pipelines with tools like Jenkins and GitLab CI.

Education:

University of Cincinnati | Cincinnati, Ohio Computer Engineering in Master of Engineering | CGPA: 4.0/4.0

08/2023 – present

- **Relevant Coursework:** Data Security and Privacy, Advanced Distributed Operating Systems, Advanced Database Management, Artificial Intelligence/Machine Learning, Natural Language Processing, Cloud Computing, Principles of Modern Networking, Advanced Algorithms.

Institute of Aeronautical Engineering | Bachelor of Technology in Computer Science & Engineering | CGPA: 8.45/10

06/2017 - 07/2021

- Published IEEE paper on ["An Efficient Text Synthetization Method by Utilizing Text Mining Techniques and Natural Language processing for the Usage of Intellectually Disabled Individuals"](#).

Personal Projects:

- **College Resource Allocation System:** developed a College Resource Allocation System using Angular, Django, TensorFlow, and PostgreSQL, seamlessly deployed with Docker. It features JWT and OAuth2 for security, ensuring efficient and secure management of college facilities.
- **Campus Event Hub (SPANDANA):** Developed an Angular and Node.js based platform, enhancing college event engagement. This resulted in a 60% increase in attendance, engaging 70% of students in its first semester.
- **SparkML-Pipeline: Scalable Real-time Data Processing Framework:** Engineered a scalable PySpark pipeline capable of processing over 500 GB of raw data, transforming it into structured Spark DataFrames and resilient distributed datasets (RDDs) for downstream processing.
- **EasyDocs - Document Manipulation & Conversion Tool:** [\[View on GitHub\]](#): developed "EasyDocs," a sophisticated document manipulation tool for students. Integrated advanced features such as automated content summarization from diverse online sources, rapid summarization of lecture notes using NLP algorithms, and OCR for extracting text from images. Enhanced user experience with the "Vishu" chatbot for navigation. Deployed functionalities for converting entire PDFs to audio with adjustable playback speeds, leveraging text-to-speech
- **Grievance App for College Campus:** Developed a Kotlin-based Android app with Firebase integration to streamline college grievance process.